RESEARCH PROGRAM · SPINOR

The Machine with No Long-Term Memory: A Level Research Program ES1–ES5

Research Program Monograph · Version 1.0.0

The machine with no long-term memory: spinor phases θ = π/2, π/3, π/7, theorems TS1–TS3 and five campaigns with pre-registered thresholds; volatility as the condition of determinism.

4/6/14	≈ 1/N	O(1)
EXACT CYCLES	POST-RESET	STEP COST

Isaev Iskhak Khamzatovich · ORCID 0009-0003-7299-0701

AB Cloud Quantum Research · DOI 10.5281/zenodo.21825394

2026 · PROPRIETARY

Table of Contents

Research Program: The Spinor Study **2**

1. Program goals . . . 2
2. Models . . . 2
3. Executed campaigns ES1–ES5 . . . 2
4. The next wave . . . 3
5. Honest boundaries . . . 3
6. Campaigns in detail . . . 3
6.1. Five measurements of volatility . . . 3
7. The full campaign table . . . 4
8. How to reproduce . . . 4
9. Further reading . . . 4

Research Program: The Spinor Study

The spinor program drives the central claim "reads, does not carry" to explicit models: the only state is the spinor phase; a step is a rotation by θ. Three theorems, five campaigns, all PASS; the capacity-versus-robustness trade-off is fixed as an honest boundary.

1. Program goals

- state and prove the properties of the volatile state: cycle closure (TS1), reset irreversibility (TS2), O(1) steps versus the Ω(t) log (TS3);
- measure the models with campaigns carrying pre-registered thresholds;
- prepare the models for processor stage P1 (memory V21).

2. Models

Table 4.2. Three models and their parameters

Model	θ	N = 2k	Bits/step	σ*	Purpose
S1	π/2	4	2.000	0.135	maximal robustness
S2	π/3	6	2.585	0.089	balance
S3	π/7	14	3.807	0.038	maximal capacity

3. Executed campaigns ES1–ES5

Table 4.3. Spinor campaigns

Campaign	Measured	Result	Verdict
ES1	cycles	periods 4/6/14 = 2k exactly; gap = step	PASS
ES2	reset	I(pre;post) = 0; reconstruction ≈ 1/N	PASS
ES3	time	step slope −1.5e−12; log ×129	PASS
ES4	task	a Markov policy is solvable without memory	PASS
ES5	noise	diffusion bounded by the silence zone	PASS

The errata ES2a and ES5a: the thresholds of campaigns ES2 and ES5 were replaced before the verdicts were accepted; the replacements are published in the verdict files. The $\log_2 N$ versus $1/k$ trade-off is a permanent boundary: capacity grows, the silence zone narrows.

4. The next wave

- phases $\theta = \pi/k$ for $k > 7$: searching for the inflection points of the trade-off;
- task compilation into phase cycles: a formal grammar of policies;
- a diffusion channel with correlations: beyond the Markov assumption.

5. Honest boundaries

Not every task is solvable without memory: the applicability class is tasks with a Markov policy. Reconstruction $\approx 1/N$ is the chance level, not "almost zero": information is erased to the limit. A log gives $\Omega(t)$ under any "smart" scheduling — this is a lower bound, not an implementation defect.

6. Campaigns in detail

6.1. Five measurements of volatility

ES1 — cycles

The first measurement campaign: the rotation periods for $\theta = \pi/2, \pi/3, \pi/7$ equal 4, 6 and 14 — exactly $N = 2k$, and the gap between distinguishable states equals the step $2\pi/N$. The volatile phase turned out to be an exact counter: no drift, no memory.

ES2 — reset and erasure

A phase reset erases everything: the pre/post mutual information is zero, reconstruction 0.2501/0.1674/0.0717 $\approx 1/N$ — the chance level. Time after the reset is defined only modulo N. Theorem TS2 gets its numerical body here; erratum ES2a recorded the threshold replacement before the verdict.

ES3 — real time

The contrast experiment: the memoryless machine holds O(1) steps (slope −1.5e−12 ms/step — zero), the eternal-log machine grows ×129 and misses deadlines. This is the numerical proof of TS3 and the main argument of the "reads, does not carry" architecture: a log is not reliability but a cause of failure.

ES4 — the memoryless task

The applicability check: a sequence of decisions inside the cycle without storage; on the class of tasks with a Markov policy the quality is comparable to a memory-bearing machine. The boundary is honestly fixed: tasks with substantial history require another level of the architecture.

ES5 — noise and the silence zone

Phase diffusion is bounded by the silence zone σ*: S1 (0.135 rad) is more robust, S3 (0.038 rad) has more capacity — the capacity/robustness trade-off is measured. Erratum ES5a: the threshold was replaced before the verdict was accepted, and the replacement is published.

7. The full campaign table

The full campaign table of the level

Campaign	Title	Key result
ES1	Phase cycles	
ES2	Volatility	
ES3	Real time	
ES4	Memoryless task	
ES5	Noise diffusion	

8. How to reproduce

```
cd research/realtime_spinor_memory
python3 experiments/run_all.py        # ES1–ES5 (~2.5 min, all PASS)
python3 make_figures.py               # 4-dimension tiles
python3 make_theorem_docs.py          # 12 TS1–TS3 documents
python3 -m pytest tests/ -q           # 11 tests
```

9. Further reading

- theory: docs/theory.en.md (EN) and docs/theory.md (RU);
- implementation: spinor/core.py — the model core, spinor/tasks.py — tasks;
- verdicts: data/es1…es5_verdict.json;
- theorems: theorems/TS1_cycle, TS2_volatility, TS3_realtime.